

Module 4 Activity 2: Writing Geometries

Purpose:

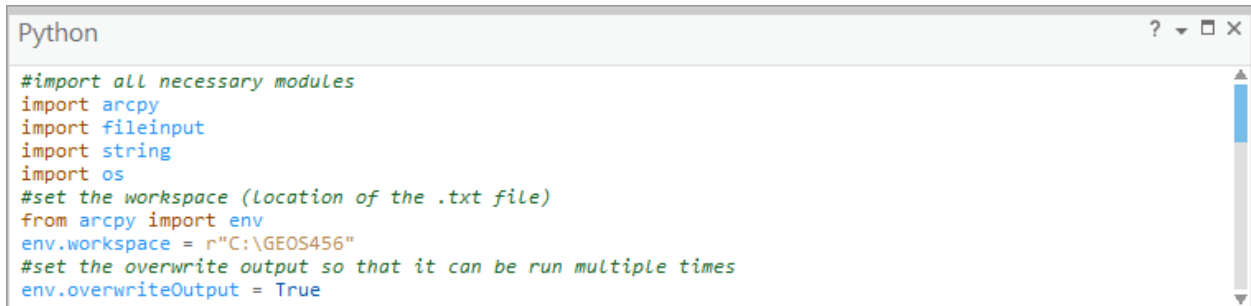
Write geometries by creating an empty feature class and using cursors to populate the empty feature class from a text file.

Instructions:

Any Python environment will work for this exercise.

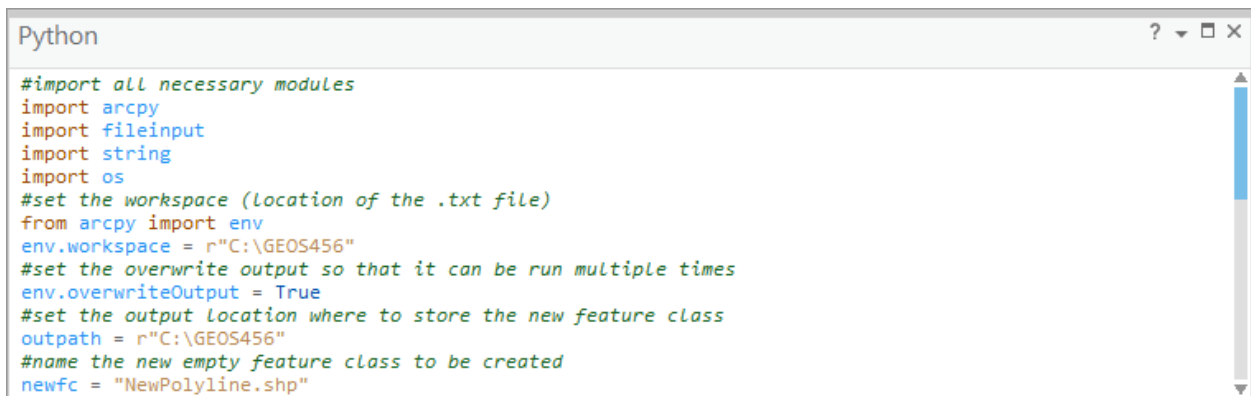
Write a polyline geomtry:

1. Import the **arcpy** module and other necessary modules
2. Set the workspace. This needs to be where the text file is located
3. Ensure that the overwrite output is set to **True**



```
Python
#import all necessary modules
import arcpy
import fileinput
import string
import os
#set the workspace (location of the .txt file)
from arcpy import env
env.workspace = r"C:\GEOS456"
#set the overwrite output so that it can be run multiple times
env.overwriteOutput = True
```

4. Set the output location where the new feature class will be stored
5. Select a name for the new shapefile to be created



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import os
#set the workspace (location of the .txt file)
from arcpy import env
env.workspace = r"C:\GEOS456"
#set the overwrite output so that it can be run multiple times
env.overwriteOutput = True
#set the output location where to store the new feature class
outpath = r"C:\GEOS456"
#name the new empty feature class to be created
newfc = "NewPolyline.shp"
```

6. Use the **CreateFeatureClass** function to create the empty feature class named above and specify the geometry type to be **Polyline**
7. Specify the text file that will be used to populate the new empty feature class with

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env.workspace = r"C:\GEOS456"
#set the overwrite output so that it can be run multiple times
env.overwriteOutput = True
#set the output location where to store the new feature class
outpath = r"C:\GEOS456"
#name the new empty feature class to be created
newfc = "NewPolyline.shp"
#use the Create Feature Class function to create the new empty feature
#with the name specified in the line above
arcpy.CreateFeatureclass_management(outpath, newfc, "Polyline")
#specify the text file to be used to populate the empty feature class with
infile = r"C:\GEOS456\coordinates.txt"
```

At this point we have created a new empty feature class called **NewPolyline**. The next steps will incorporate an insert cursor that will insert the geometry objects into the new polyline. These geometry object will need to be stored in an Array. Once the **InsertCursor** parameters are populated, the **fileinput** module will read the text file and the **split** method will return a list of strings with a delimiter. With no set arguments in the **split** method, it will use the white space as the delimiter and assign each list to **ID**, **X** and **Y**.

8. Specify the geometry object **SHAPE@** inside the cursor
9. Store the geometry object in an array

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#set the overwrite output so that it can be run multiple times
env.overwriteOutput = True
#set the output location where to store the new feature class
outpath = r"C:\GEOS456"
#name the new empty feature class to be created
newfc = "NewPolyline.shp"
#use the Create Feature Class function to create the new empty feature
#with the name specified in the line above
arcpy.CreateFeatureclass_management(outpath, newfc, "Polyline")
#specify the text file to be used to populate the empty feature class with
infile = r"C:\GEOS456\coordinates.txt"
#incorporate the Insert Cursor and specify the geometry object
cursor = arcpy.da.InsertCursor(newfc, ["SHAPE@"])
#store the SHAPE@ geometry object in an Array
array = arcpy.Array()
```

Once the geometry object is stored in the array, the array will assign the values to an XY location

10. Begin a **for** loop
11. Use the **fileinput** module to read the text file
12. Use the **split** method to use the white space as a delimiter
13. Assign the values stored in the array to the XY locations from the text file to the new feature class
14. Use the **insertRow** to insert the values to the feature class

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infile = r"C:\GEOS456\coordinates.txt"
#incorporate the Insert Cursor and specify the geometry object
cursor = arcpy.da.InsertCursor(newfc, ["SHAPE@"])
#store the SHAPE@ geometry object in an Array
array = arcpy.Array()
#start a for loop
#the fileinput module will read the text file
for line in fileinput.input(infile):
    #the split method will use the white space as a delimiter
    ID, X, Y = str.split(line, " ")
    #the Array will assign the stored values to the polyline
    array.add(arcpy.Point(X, Y))
#the insert row will add the record to the line
cursor.insertRow([arcpy.Polyline(array)])
```

15. Close the **fileinput** reader
16. Delete the cursor object to remove any lock on the data

```
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import fileinput
import string
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from arcpy import env
env.workspace = r"C:\GEOS456"
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#incorporate the Insert Cursor and specify the geometry object
cursor = arcpy.da.InsertCursor(newfc, ["SHAPE@"])
#store the SHAPE@ geometry object in an Array
array = arcpy.Array()
#start a for loop
#the fileinput module will read the text file
for line in fileinput.input(infile):
    #the split method will use the white space as a delimiter
    ID, X, Y = str.split(line, " ")
    #the Array will assign the stored values to the polyline
    array.add(arcpy.Point(X, Y))
#the insert row will add the record to the line
cursor.insertRow([arcpy.Polyline(array)])
#close the text file reader
fileinput.close()
#delete the cursor to remove any locks on the data
del cursor
```

17. Run the script

18. Examine your results

You should see a new feature class called **NewPolyline** created in your workspace. You can open this new feature class in ArcMap or ArcCatalog. The shape of the line is determined by the values from the text file and are the vertices of that new line. Remember that you cannot see the locations of the vertices in the attribute table, which is why you only see one attribute record, nor are the XY values listed anywhere in the attributes.

This new feature class has no specified coordinate system, so it will be unprojected.

